

CBCS SCHEME

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Second Semester M.Tech. Degree Examination, June/July 2024 Big Data Analytics

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Discuss the dimensionalities of HDFS design.		6	L2	CO1
	b.	With neat diagram, explain MapReduce dataflow to illustrate the task of finding the maximum temperature for every year using the NCDC data.		8	L3	CO2
	c.	Differentiate between Relational database and Schemaless database.		6	L3	CO2
OR						
Q.2	a.	Write a note on following : i) HDFS federation ii) HDFS high availability.		8	L2	CO2
	b.	Describe the anatomy of file. Write in HDFS.		6	L2	CO2
	c.	Compare Grid computing and Volunteer computing.		6	L3	CO1
Module – 2						
Q.3	a.	Summarize the anatomy of YARN application run, with neat diagram.		8	L2	CO2
	b.	Outline the limitations of MapReduce that are overcome by YARN.		6	L2	CO2
	c.	Demonstrate the role of delay scheduling to increase efficiency of the cluster.		6	L3	CO2
OR						
Q.4	a.	Infer how data integrity is ensured in HDFS.		8	L2	CO2
	b.	Illustrate the process of serialization and deserialization in Hadoop. What is desired RPC serialization format?		6	L2	CO2
	c.	With neat diagram describe the internal structure of sequence file, with block compression.		6	L2	CO2
Module – 3						
Q.5	a.	Explain how Hadoop runs a MapReduce job with neat diagram.		10	L2	CO4
	b.	Given the problem to count the number of words in a text file, illustrate MapReduce shuffle and sort steps in detail, with neat diagram.		10	L3	CO4
OR						

Q.6	a.	Interpret the different types of failures in MapReduce.	10	L2	CO1
	b.	Demonstrate the need for speculative execution for optimization of MapReduce model. Mention its properties that tasks can enable and disable.	10	L3	CO1

Module – 4

Q.7	a.	Classify and explain the following : i) File input format ii) Sequence file input format.	8	L2	CO1
	b.	Describe Text output and Binary output formats.	6	L2	CO3
	c.	Explain how Flume uses transactions to guarantee delivery.	6	L2	CO1

OR

Q.8	a.	Illustrate Flume configuration using spooling diagram and logger sink with neat diagram.	8	L2	CO1
	b.	Describe how aggregating flume is achieved.	6	L2	CO1
	c.	Explain the role of multiple sinks for load balancing or failover.	6	L2	CO1

Module – 5

Q.9	a.	Illustrate different types of execution for PIG.	6	L2	CO1
	b.	Mention PIG latin Relational operators.	8	L3	CO1
	c.	Outline the concept of functions for Pig.	6	L2	CO1

OR

Q.10	a.	Interpret the need for Apache Spark for large scale data processing.	6	L2	CO3
	b.	Write a note on three ways to create RDDs.	9	L2	CO3
	c.	What are the different aspects of serialization in Spark?	5	L2	CO3